

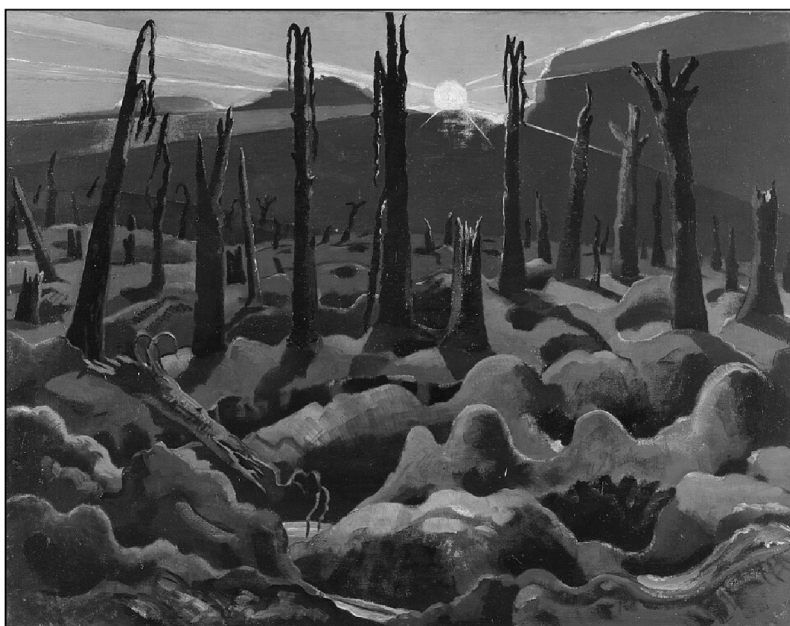
THE HISTORY DEPARTMENT

GCSE HISTORY (9-1) • PAPER 1: THEMATIC STUDY & HISTORIC ENVIRONMENT

SECTION A: HISTORIC ENVIRONMENT • 1914–1918

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Injuries, Treatment and the Trenches • 16-Page Master Study & Exam Workbook



RAMC stretcher bearers carrying a severely wounded soldier through liquefied Flanders clay at Passchendaele, 1917 (Imperial War Museum Q 2757).

EDEXCEL PAPER 1 SECTION A SPECIFICATION MODULES (100% EXAM MASTERY)

- | | |
|---|---|
| • KT5.1: Trench System, Geography & Battles (Ypres, Somme, Arras, Cambrai) | • KT5.4: Chain of Evacuation (Bearers, RAP, ADS/MDS, CCS, Base Hospitals) |
| • KT5.2: Trench Environment (Trench Foot, Whale Oil, Lice, Fever, Dysentery) | • KT5.5: Surgical Breakthroughs (Thomas Splint, Debridement, Carrel-Dakin, X-Rays) |
| • KT5.3: Combat Trauma (High-Explosive Shrapnel, Gas Gangrene, Gas Helmets) | • KT5.6: Lifesaving Advances (Robertson Blood Depot, Cushing, Gillies Plastic Surgery) |

Candidate Name: _____

Target Grade: _____ Teacher / Class: _____

Exam Rules: Section A accounts for 16 of the 52 marks on Paper 1 (Q1 Feature 4m + Q2a Utility 8m + Q2b Follow-up 4m).



SCAN FOR AUDIO & INTERACTIVE MODELS

WESTERN FRONT LIVING TIMELINE: 1914–1918

Dual-Coding Synthesis: Complete all 6 chronological visual/textual missions

1. OCT–NOV 1914: THE FIRST BATTLE OF YPRES & STATIC TRENCH WARFARE

KT5.1

The mobile war ends; armies dig in from the Channel to Switzerland. British defend the low-lying Ypres Salient against German ridges.

[SKETCH: Ypres Salient Bulge + Traverses]

Exam Synthesis: Why were frontline trenches dug in zig-zag traverses rather than straight lines?

2. 1915: CHLORINE GAS (2ND YPRES) & THE BRODIE STEEL HELMET

KT5.2 • KT5.3

First gas attack at 2nd Ypres (April 1915); troops improvise urine pads then PH helmets. John Brodie patents the pressed-steel helmet.

[SKETCH: Brodie Helmet vs Shrapnel]

Exam Synthesis: How did the Brodie helmet slash fatal head wounds by an estimated 80%?

3. DEC 1915: ROBERT JONES INTRODUCES THE THOMAS SPLINT

KT5.5

Robert Jones deploys Hugh Owen Thomas's splint to frontline aid posts, applying mechanical traction to broken femurs.

[SKETCH: Thomas Splint Groin Ring & Traction]

Exam Synthesis: Why did continuous traction reduce compound fracture mortality from 80% to 20%?

4. JULY 1916: THE BATTLE OF THE SOMME & WOUND DEBRIDEMENT

KT5.1 • KT5.5

57,470 casualties on Day 1; CCSs overwhelmed. Surgeons master wound excision (debridement) and Carrel-Dakin bleach irrigation.

[SKETCH: Carrel-Dakin Perforated Irrigation Tubes]

Exam Synthesis: Why did Carrel-Dakin solution have to be freshly made in labs every 6 hours?

5. 1917: ARRAS CHALK HOSPITAL & PASSCHENDAELE MUD CRISIS

KT5.1 • KT5.4

Thompson's Cave under Arras shelters 700 hospital beds with electric lights; 3rd Ypres mud requires 6 bearers per stretcher.

[SKETCH: Thompson's Cave Subterranean Hospital]

Exam Synthesis: How did the chalk geology of Arras contrast with the clay water table of Ypres?

6. NOV 1917: OSWALD ROBERTSON'S BLOOD BANK & SIDCUR PLASTIC SURGERY

KT5.6

Robertson uses citrate-glucose blood stored on ice at Cambrai; Gillies pioneers tubed pedicles at Queen's Hospital Sidcup.

[SKETCH: Robertson Ice Chest / Gillies Tubed Pedicle]

Exam Synthesis: Why was citrate-glucose essential for keeping stored blood viable for up to 28 days?

THE WESTERN FRONT EVACUATION CHAIN

RAMC & FANY Lifeline: From No Man's Land to Base Hospital

THE 6 SEQUENTIAL STATIONS OF THE EVACUATION CHAIN		
1. STRETCHER BEARERS Frontline & No Man's Land • 16 men per battalion of 1,000. • 4 to 6 bearers per stretcher in mud. • Apply first field dressing under fire.	2. REGIMENTAL AID POST (RAP) 200–300 yards behind front line • Located in communication trench/cellar. • RMO gives morphine & tetanus serum. • <i>Cannot perform major surgery.</i>	3. DRESSING STATIONS (ADS/MDS) 1 to 3 miles behind front line • Run by RAMC Field Ambulance. • Treat shock with sweet tea & blankets. • Sorts casualties for motor ambulance.
4. CASUALTY CLEARING STATION 7 to 12 miles back (rail/canal) • Surgical core: operating theatres & X-rays. • Female nurses (QAIMNS). • Triage: Walking, Immediate, Moribund.	5. BASE HOSPITALS Coastal ports (Boulogne/Calais) • Large general/stationary hospitals. • Extended recovery & specialized care. • Linked by hospital trains & canal barges.	6. "BLIGHTY" (BRITAIN) Hospital ships across Channel • Queen's Hospital Sidcup (Plastic surgery). • Roehampton (Amputee limb fitting). • Netley Hospital (Military psychiatry).

RAMC TRIAGE CLASSIFICATION AT CCS	TRANSPORT METHODS: HORSE, MOTOR & CANAL
• WHITE TAG (Walking Wounded): Minor shrapnel cuts, superficial gas irritation. Cleaned, patched, and returned to the line within days. • RED TAG (Immediate Surgery): Compound fractures, abdominal hemorrhages, chest trauma with a viable chance of survival if operated on at once. • GREEN/NO TAG (The Moribund): Hopeless wounds (severe brain destruction, massive blood loss). Made comfortable with morphine until death.	• Horse Ambulances (1914): Slow, violent jolting over shell craters caused hemorrhagic shock and re-fractured bones. • Motor Ambulances & FANY (1915+): Faster transport; voluntary women drivers braved artillery to evacuate men from dressing stations. • RAMC Canal Barges: Smooth, vibration-free water transit along French canals, ideal for post-operative brain and abdominal trauma.

FORENSIC PATIENT JOURNEY: PRIVATE THOMAS HENDERSON (2ND ROYAL SCOTS)		YPRES SECTOR • 14 SEPTEMBER 1917
08:15 Frontline: Struck by high-explosive shell shrapnel in trench. Shattered right femur (compound fracture); severe arterial bleed.	08:45 Stretcher Retrieval: 4 stretcher bearers carry him through flooded saps to the Regimental Aid Post dugout.	
09:10 RAP First Aid: RMO fits Thomas Splint with windlass traction, injects 500 units anti-tetanus serum and 1/3 grain morphine.	11:30 ADS to CCS: Transported by FANY motor ambulance van 8 miles back to No. 46 Casualty Clearing Station.	
13:00 CCS Triage: Classified Red Tag. Mobile X-ray locates 3 metal fragments. Surgeon debrides necrotic flesh; Carrel-Dakin tubes inserted.	Day 4 Base Hospital: Evacuated by heated RAMC canal barge to Boulogne; loaded on hospital ship for recovery in Blighty.	

KT5.1: THE THEATRE OF WAR: THE BRITISH SECTOR, TRENCH GEOGRAPHY & BATTLES (1914–1918)

Specification Focus: The British sector of the Western Front: theatre of war and trench system (frontline, support, reserve, communication trenches); terrain, saps, and battleground sectors (Ypres, Somme, Arras, Cambrai).

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH] MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of aseptic surgical techniques developed before 1914. [2 marks]

Feature: _____

Detail: _____

2. Describe one feature of Wilhelm Röntgen's discovery of X-rays in 1895. [2 marks]

Feature: _____

Detail: _____

3. Describe one feature of Karl Landsteiner's discovery of human blood groups in 1901. [2 marks]

Feature: _____

Detail: _____

4. Describe one feature of Florence Nightingale's pavilion hospital design in the late 19th century. [2 marks]

Feature: _____

Detail: _____

5. Describe one feature of the pre-war limitation of blood transfusions in 1914. [2 marks]

Feature: _____

Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS

- **Frontline Trench:** The foremost defensive trench situated closest to enemy lines, featuring a firestep, sandbag parapet, and barbed wire entanglements, where infantry mounted guard.
- **Traverse:** A sharp 90-degree bend or earthwork buttress constructed into trench lines to prevent enemy riflemen firing down the ditch and contain artillery shell bursts.

- **Communication Trench:** Deep, connecting trenches dug at right angles to parallel lines, allowing fresh troops, ammunition, food, and stretcher-borne casualties to move under cover.
- **Ypres Salient:** A highly vulnerable, low-lying British military bulge around the medieval town of Ypres, surrounded on three sides by German forces holding higher commanding ridges.

SOURCE A: AERIAL RECONNAISSANCE PHOTOGRAPH OF BRITISH TRENCH SYSTEM NEAR YPRES (1917) SOURCE-A-AERIAL-TRENCH-YPRES



Provenance: Official military aerial photograph taken by the Royal Flying Corps, British Sector near Ypres, 1917. Imperial War Museum Collection (Shelfmark: Q 42231).

Context: An authentic aerial reconnaissance photograph taken by the Royal Flying Corps in 1917 showing the complex defensive network of British frontline, support, and communication trenches in the Ypres sector, displaying the characteristic zig-zag design and dense shell cratering.

Royal Flying Corps aerial photograph showing the intricate zig-zag traverses of frontline, support, and communication trenches near Ypres, criss-crossed with artillery shell craters.

Hinge Question: Study Source A. How does the zig-zag design of the trenches shown in the photograph explain how British defensive engineers sought to protect soldiers from artillery shrapnel and machine-gun fire?

SOURCE B: PHOTOGRAPH OF A FLOODED COMMUNICATION TRENCH IN THE YPRES SALIENT (1917)

SOURCE-B-FLOODED-TRENCH



British soldiers navigating a communication trench submerged in muddy water during the Third Battle of Ypres (Passchendaele), showing duckboards floating and collapsed trench revetments.

Provenance: Official British Army photograph taken by Lieutenant Ernest Brooks, Western Front, Flanders, 1917. Imperial War Museum Collection (Shelfmark: Q 5935).

Context: Official photograph taken by British war photographer Ernest Brooks in 1917 during the Third Battle of Ypres, documenting the extreme waterlogging of communication trenches caused by heavy autumn rainfall and the destruction of Flanders clay drainage systems by artillery fire.

Hinge Question: How does the physical state of the trench in Source B illustrate why carrying wounded soldiers by stretcher to dressing stations frequently took up to ten hours?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source A on page 4. How useful is Source A for an enquiry into the defensive layout and design of the trench system on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, and what ground-level details does it omit?

Sentence Stems: Source A is useful for an enquiry into trench layout because it shows... • In particular, the zig-zag traverse pattern was designed to... • However, as an aerial photograph, it is limited because...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE B

How could you follow up **Source B** to find out more about treatments or conditions on the Western Front?

Detail in Source B that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

KT5.2: THE TRENCH ENVIRONMENT: MUD, VERMIN & NON-COMBAT ILLNESSES (1914–1918)

Specification Focus: Ill health arising from the trench environment: trench foot (pathology, prevention, whale oil); trench fever (body lice, delousing); dysentery (water chlorination, chloride of lime, latrines); underground shelters.

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH] MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of the communications trenches in the Western Front trench system. [2 marks]

Feature: _____ Detail: _____

2. Describe one feature of the Ypres Salient as a military battleground. [2 marks]

Feature: _____ Detail: _____

3. Describe one feature of the underground tunnel system constructed at Arras in 1916–17. [2 marks]

Feature: _____ Detail: _____

4. Describe one feature of the zig-zag traverse design used in frontline trenches. [2 marks]

Feature: _____ Detail: _____

5. Describe one feature of the Battle of the Somme (1916) for British medical services. [2 marks]

Feature: _____ Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS

• **Trench Foot:** A painful, necrotic condition of the lower feet caused by prolonged exposure to cold, wet mud and poor circulation, which could degenerate into gangrene requiring amputation.

• **Trench Fever:** A debilitating bacterial infection (Pyrexia of Unknown Origin) caused by Bartonella quintana and transmitted by body lice faeces, producing severe shin pains and high fevers.

• **Whale Oil:** A thick, pungent animal grease supplied in large barrels to frontline British troops, rubbed into bare feet daily to create a water-resistant protective barrier.

• **Body Lice:** Small wingless parasitic insects that nested in the warm seams of woollen service uniforms, biting human skin and spreading trench fever across the Western Front.

SOURCE A: SOLDIERS OF THE CHESHIRE REGIMENT GREASING FEET IN A RESERVE TRENCH (1916)

SOURCE A-CHESHIRE-REGIMENT-TRENCH



British infantrymen of the 8th Battalion, Cheshire Regiment, seated on the firestep of a trench, removing their boots and rubbing whale oil onto their feet under the supervision of a non-commissioned officer.

Provenance: Official British War Office photograph taken on the Western Front, 1916. Imperial War Museum Collection (Shelfmark: Q 5156).

Context: An authentic wartime photograph taken in 1916 showing British infantrymen of the Cheshire Regiment carrying out the mandatory daily foot inspection and greasing routine using whale oil to prevent the onset of trench foot in cold, damp conditions.

Hinge Question: Study Source A. How does this photograph demonstrate the organized preventive measures enforced by the British Army to stop soldiers from developing trench foot?

SOURCE B: PHOTOGRAPH OF THE RAMPARTS UNDERGROUND MEDICAL DRESSING STATION, YPRES (C.1916)

SOURCE-B-RAMPARTS-WW1



RAMC medical orderlies and stretcher bearers outside the subterranean vaulted brick entrance of the Ramparts Dressing Station, built deep into the 17th-century fortifications of Ypres.

Provenance: Official British military photograph taken in the Ypres sector, c.1916. Imperial War Museum Collection (Shelfmark: Q 10762).

Context: Photograph taken c.1916 showing the entrance to the Ramparts Advanced Dressing Station in Ypres. Carved deep into the massive 17th-century Vauban brick earthworks near the Lille Gate, it provided a bomb-proof, dry, subterranean refuge where medical staff could treat wounded men and sick soldiers free from direct artillery shell fire.

Hinge Question: How does the underground construction of the Ramparts Dressing Station in Source B explain how the RAMC adapted existing architecture to protect casualties from artillery fire and wet weather?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source A on page 6. How useful is Source A for an enquiry into the methods used by the British Army to prevent trench foot on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an official British War Office photograph taken in 1916 showing the Cheshire Regiment. What does it reveal about supervised foot inspection routines, and why might an official photograph show clean conditions?

Sentence Stems: Source A is useful for investigating the prevention of trench foot because it depicts... • This corroborates historical knowledge that soldiers were required to... • However, the source is limited because it is an official photograph showing...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE B

How could you follow up **Source B** to find out more about treatments or conditions on the Western Front?

Detail in Source B that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

KT5.3: BATTLEFIELD TRAUMA: HIGH EXPLOSIVE SHRAPNEL, GAS ATTACKS & INFECTION (1914–1918)

Specification Focus: Wounds, injuries and diseases: high-explosive artillery shells, shrapnel fragments; infection (gas gangrene, *Clostridium welchii*, tetanus); head trauma and the Brodie helmet; nature of chemical gas attacks (chlorine, phosgene, mustard gas) and respirators.

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH] MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of the medical condition known as trench foot. [2 marks]

Feature: _____ Detail: _____

2. Describe one feature of the preventive routine enforced by officers against trench foot. [2 marks]

Feature: _____ Detail: _____

3. Describe one feature of trench fever (Pyrexia of Unknown Origin). [2 marks]

Feature: _____ Detail: _____

4. Describe one feature of the transmission of trench fever. [2 marks]

Feature: _____ Detail: _____

5. Describe one feature of water sanitation measures taken to prevent dysentery in the trenches. [2 marks]

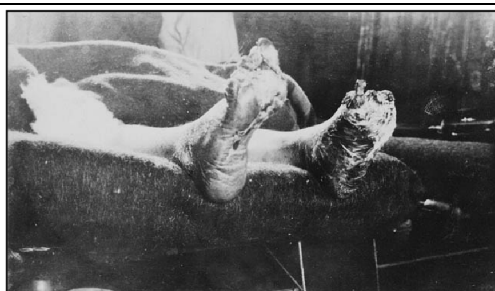
Feature: _____ Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS

- **High-Explosive Shell:** Heavy artillery ammunition filled with TNT that detonated on impact, shattering its steel casing into thousands of jagged, high-velocity shrapnel fragments.
- **Tetanus:** A lethal bacterial disease (*Clostridium tetani*) from soil causing severe muscle spasms and lockjaw, successfully prevented by routine injections of tetanus antitoxin.
- **Gas Gangrene:** A ferocious, life-threatening infection caused by anaerobic bacteria (*Clostridium welchii*) from manured soil that produced gas bubbles and rotted living muscle tissue.
- **Brodie Helmet:** A pressed-steel combat helmet patented in 1915 by John Brodie, featuring a wide brim to protect soldiers' heads from downward-falling artillery shrapnel.

SOURCE A: CLINICAL MEDICAL RECORD PHOTOGRAPH OF SEVERE GANGRENE IN LOWER LIMB (C.1916)

SOURCE-A-TRENCH-FOOT-CLINICAL



RAMC clinical photograph documenting severe necrotic tissue damage and secondary gangrenous infection in the foot of a British soldier admitted to a base hospital from the frontline trenches.

Provenance: Official clinical photograph produced for the British Medical Research Committee / RAMC Case Records, c.1916. Wellcome Collection, London (Shelfmark: RAMC 442/1).

Context: An official clinical photograph compiled for the Royal Army Medical Corps teaching archives c.1916, documenting the devastating progression of anaerobic necrosis and moist gangrene in a soldier's lower limb following prolonged exposure to contaminated Flanders mud and untreated tissue trauma.

Hinge Question: Study Source A. How does this clinical photograph illustrate the severity of anaerobic bacterial infections like gangrene when soil bacteria entered damaged, devitalized human flesh?

SOURCE B: BRITISH ARMY PHENATE-HEXAMINE (PH) ANTI-GAS HELMET (1915–1916)

SOURCE-B-BRITISH-PH-GAS-HELMET



An authentic British PH gas helmet manufactured in 1915, featuring chemically treated flannel cloth, mica/glass eyepieces, and a rubberized exhale valve to protect soldiers from chlorine and phosgene gas.

Provenance: Primary military artifact preserved in the Imperial War Museum Collection, London (Accession: EPH 4220). Manufactured by the British War Office, 1915–1916.

Context: An authentic surviving example of the British Phenate-Hexamine (PH) anti-gas helmet introduced in late 1915. Made of double-layered flannel impregnated with sodium phenate and hexamine to chemically neutralize chlorine and phosgene gas, it was tucked into the collar of the uniform tunic.

Hinge Question: How does the design of the PH helmet in Source B reflect both the rapid improvisation of the British Army against chemical weapons and its physical limitations compared to later respirators?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source B on page 8. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) helmet manufactured in 1915–1916. What physical chemical protection features does it display, and what combat limitations did flannel cloth helmets possess?

Sentence Stems: Source B is useful for investigating gas protection because it provides physical evidence of... • The chemically treated flannel was designed to neutralize... • However, as a standalone artifact, Source B does not show...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE A

How could you follow up **Source A** to find out more about treatments or conditions on the Western Front?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

KT5.4: THE CHAIN OF EVACUATION: STRETCHER BEARERS, RAP, DRESSING STATIONS & CCS (1914–1918)

Specification Focus: The work of the RAMC and FANY: the Chain of Evacuation (stretcher bearers, Regimental Aid Posts, Field Ambulances and Dressing Stations, Casualty Clearing Stations, Base Hospitals); triage systems; transport methods (motor ambulances, ambulance trains, canal barges).

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH] MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of the wounds caused by high-explosive artillery shells on the Western Front. [2 marks]

Feature: _____ Detail: _____

2. Describe one feature of gas gangrene in wounded soldiers. [2 marks]

Feature: _____ Detail: _____

3. Describe one feature of the Brodie steel helmet introduced in 1915. [2 marks]

Feature: _____ Detail: _____

4. Describe one feature of the first German chlorine gas attack at the Second Battle of Ypres in April 1915. [2 marks]

Feature: _____ Detail: _____

5. Describe one feature of the British Small Box Respirator (SBR) issued from 1916. [2 marks]

Feature: _____ Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS

- **Chain of Evacuation:** The organized, multi-stage medical evacuation network designed to transport wounded soldiers from the frontline firestep back to specialist hospitals in Britain.
- **Casualty Clearing Station (CCS):** The critical surgical field hospital situated 7–12 miles behind the front line beside rail networks, equipped with sterile operating theatres, X-rays, and female nurses.

- **Regimental Aid Post (RAP):** The foremost medical dugout located within 200–300 yards of the frontline, where the medical officer administered field dressings, morphine, and tetanus antitoxin.
- **Triage:** The systematic sorting of casualties into three categories upon arrival at a CCS (walking wounded, immediate surgery, and moribund) to prioritize life-saving operations.

SOURCE A: RAMC STRETCHER BEARERS CARRYING A CASUALTY AT PASSCHENDAELE (1917)

SOURCE-A-STRETCHER-BEARERS-PASSCHENDAELE



Four British stretcher bearers struggling through deep, liquid mud and waterlogged shell craters to evacuate a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

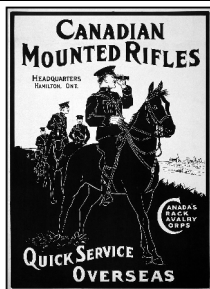
Provenance: Official British Army photograph taken by Lieutenant Ernest Brooks on the Western Front, Flanders, 1917. Imperial War Museum Collection (Shelfmark: Q 2757).

Context: Official photograph taken by British war photographer Ernest Brooks in 1917 during the Third Battle of Ypres (Passchendaele), capturing the immense physical difficulty experienced by RAMC stretcher bearers navigating devastated, liquid-mud battlefields under continuous fire.

Hinge Question: Study Source A. How does this photograph illustrate why transporting wounded soldiers from the frontline to dressing stations took many exhausting hours in the Ypres sector?

SOURCE B: CONVERTED RAMC AMBULANCE CANAL BARGE ON FRENCH WATERWAYS (C.1916)

SOURCE-B-RAMC-CANAL-BARGE



Provenance: Official British military photograph taken in northern France, c.1916. Imperial War Museum Collection (Shelfmark: Q 11048).

Context: An official photograph taken c.1916 showing an RAMC ambulance canal barge in northern France. Converted barges carried up to thirty stretcher cases along calm inland waterways, providing a vibration-free, smooth evacuation method for soldiers recovering from delicate abdominal surgery or fractured skulls who could not survive the jolting of road ambulances.

A French canal barge converted by the Royal Army Medical Corps into an ambulance barge, fitted with canvas awnings, clean hospital cots, and heating, transporting fragile post-operative casualties to Base Hospitals.

Hinge Question: How does the use of converted canal barges shown in Source B explain how the RAMC adapted natural infrastructure to improve casualty survival during transit to Base Hospitals?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source A on page 10. How useful is Source A for an enquiry into the difficulties of evacuating wounded soldiers from the battlefield on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an official British Army photograph taken by Lieutenant Ernest Brooks during the Third Battle of Ypres in 1917. What does it reveal about the physical mud and stretcher logistics, and what combat dangers are invisible?

Sentence Stems: Source A is useful for investigating evacuation difficulties because it shows... • In the liquid mud of Passchendaele, carrying one stretcher required... • However, as an isolated photograph, Source A cannot convey...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE B

How could you follow up **Source B** to find out more about treatments or conditions on the Western Front?

Detail in Source B that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

KT5.5: SURGICAL BREAKTHROUGHS: THE THOMAS SPLINT, WOUND DEBRIDEMENT & MOBILE X-RAYS (1914–1918)

Specification Focus: Medical advances on the Western Front: the Thomas Splint (Hugh Owen Thomas, Robert Jones) reducing compound femur mortality; wound debridement and delayed primary closure; the Carrel-Dakin antiseptic irrigation method; mobile X-ray units and radiology.

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH]
MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of the Regimental Aid Post (RAP) in the Chain of Evacuation. [2 marks]

Feature: _____

Detail: _____

2. Describe one feature of the role of stretcher bearers on the Western Front. [2 marks]

Feature: _____

Detail: _____

3. Describe one feature of the Casualty Clearing Station (CCS) in the Chain of Evacuation. [2 marks]

Feature: _____

Detail: _____

4. Describe one feature of the triage system operated at Casualty Clearing Stations. [2 marks]

Feature: _____

Detail: _____

5. Describe one feature of the transport methods used between Casualty Clearing Stations and Base Hospitals. [2 marks]

Feature: _____

Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS	
Compound Fracture: A severe bone fracture where jagged broken bone ends pierce through muscle and skin, creating an open wound highly vulnerable to lethal bacterial infection. Traction: The steady mechanical pulling force applied to a fractured limb to counteract violent muscle spasms, pull bone ends into alignment, and prevent internal hemorrhaging.	Thomas Splint: A rigid metal frame designed by Hugh Owen Thomas and championed by Robert Jones, applying traction to broken thigh bones and reducing femur mortality from 80% to 20%. Wound Debridement: The surgical cutting away (excision) of all dead, devitalized tissue and foreign matter around a wound to starve anaerobic bacteria of the material they need to cause gangrene.

SOURCE A: PHOTOGRAPH OF THE THOMAS SPLINT APPLIED TO A WOUNDED BRITISH SOLDIER (C.1916)

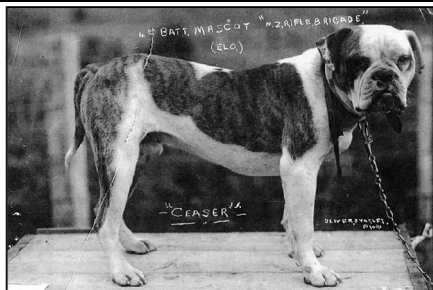
Provenance: Official photograph produced for the RAMC Training and Photographic Archive, Western Front, c.1916. Wellcome Collection, London (Shelfmark: RAMC 512/3).

Context: An official instructional photograph compiled for the Royal Army Medical Corps c.1916, illustrating the precise method of applying the Thomas Splint to a patient with a compound fracture of the femur. The padded ring fits firmly against the pelvis while traction is maintained on the foot, pulling the limb straight to prevent movement of broken bone fragments.

SOURCE-A-THOMAS-SPLINT-AUTHENTIC

Hinge Question: Study Source A. How does the mechanical design of the Thomas Splint shown in the photograph explain why survival rates for fractured femurs rose from 20% in 1914 to 80% by 1917?

The Thomas Splint: Before Robert Jones introduced it, an 80% chance of death; after Jones, an 80% chance of living to complain about the food.
12/16

SOURCE B: PHOTOGRAPH OF AN OPERATING THEATRE AT AN RAMC CASUALTY CLEARING STATION (1917)**SOURCE-B-CASUALTY-CLEARING-STATION-WW1**

Surgeons and QAIMNS nurses performing emergency operations inside a large canvas tented operating theatre at No. 47 Casualty Clearing Station, showing sterile operating tables, autoclaves, and electric lighting.

Provenance: Official British War Office photograph taken on the Western Front, 1917. Imperial War Museum Collection (Shelfmark: Q 6185).

Context: Official British Army photograph taken in 1917 showing the operating theatre of an active Casualty Clearing Station in northern France. Operating around the clock during major offensives, surgical teams worked at multiple sterile tables under intense electric lighting to debride contaminated wounds, perform emergency amputations, and irrigate tissue.

Hinge Question: How does the operating environment shown in Source B illustrate the transition of Casualty Clearing Stations from simple triage huts into advanced surgical hospitals?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source A on page 12. How useful is Source A for an enquiry into treatments for wounded soldiers with fractured bones on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an instructional training photograph produced for the RAMC archive c.1916. What does it demonstrate about mechanical traction, and why might a staged instructional image differ from emergency frontline practice?

Sentence Stems: Source A is useful for investigating fracture treatments because it illustrates... • The mechanical traction applied by the splint prevented... • However, being an instructional training photo in a clean ward, it does not reflect...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE B

How could you follow up **Source B** to find out more about treatments or conditions on the Western Front?

Detail in Source B that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

KT5.6: LIFESAVING INNOVATIONS: BLOOD STORAGE, BRAIN SURGERY & PLASTIC RECONSTRUCTION (1914–1918)

Specification Focus: Medical advances on the Western Front: blood transfusions and storage (Landsteiner, Hustin, Rous and Turner, Captain Oswald Robertson and the Cambrai blood bank); specialized neurosurgery (Harvey Cushing); plastic and facial reconstruction (Harold Gillies, Queen's Hospital Sidcup, tubed pedicle).

EDEXCEL PAPER 1 Q1 RETRIEVAL DRILLS: 5X "DESCRIBE ONE FEATURE OF..." [2 MARKS EACH] MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL

1. Describe one feature of the Thomas Splint introduced to the Western Front in December 1915. [2 marks]

Feature: _____ Detail: _____

2. Describe one feature of the surgical technique known as wound debridement. [2 marks]

Feature: _____ Detail: _____

3. Describe one feature of the Carrel-Dakin method of treating infected battlefield wounds. [2 marks]

Feature: _____ Detail: _____

4. Describe one feature of mobile X-ray units deployed by the Royal Army Medical Corps. [2 marks]

Feature: _____ Detail: _____

5. Describe one feature of the delayed primary closure technique developed by wartime surgeons. [2 marks]

Feature: _____ Detail: _____

KEY HISTORICAL VOCABULARY & FORENSIC DEFINITIONS

- **Sodium Citrate:** A chemical anticoagulant added to fresh blood in 1914 by Hustin and Agote that stopped blood from clotting outside the body by binding calcium ions.
- **Harvey Cushing:** Pioneering American neurosurgeon who operated under local anesthetic and used electromagnets to extract shrapnel from brain tissue, reducing head wound mortality to 29%.

- **Blood Depot:** The world's first blood bank established by Captain Oswald Robertson at the Battle of Cambrai (1917), storing citrated blood in ice chests for up to 28 days.
- **Harold Gillies:** New Zealand-born surgeon who founded Queen's Hospital, Sidcup, pioneering modern plastic surgery and the tubed pedicle skin flap for disfigured soldiers.

SOURCE A: CAPTAIN OSWALD ROBERTSON'S BLOOD STORAGE AND TRANSFUSION APPARATUS AT CAMBRAI (1917)

SOURCE-A-ROBERTSON-BLOOD-DEPOT-1917



The authentic wooden ice chest, glass collection flasks, rubber delivery tubing, and sodium citrate solutions utilized by Captain Oswald Robertson to establish the world's first blood bank during the Battle of Cambrai in November 1917.

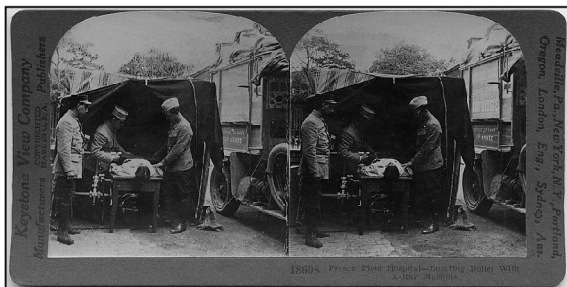
Provenance: Primary medical artifact preserved in the Royal Army Medical Corps Museum / Wellcome Historical Medical Collection (Accession: RAMC 628/1).

Context: An authentic surviving artifact photograph of Captain Oswald Robertson's blood storage and transfusion kit used at the Battle of Cambrai in November 1917. Robertson collected blood in sterile glass bottles containing sodium citrate and dextrose (glucose) and stored them inside an insulated wooden ice chest, enabling stored blood to be transfused up to 26 days after collection.

Hinge Question: Study Source A. How does this blood storage apparatus explain how medical officers at the Battle of Cambrai were able to treat soldiers suffering from severe shock before operations began?

SOURCE B: PHOTOGRAPH OF A MOBILE FIELD HOSPITAL X-RAY UNIT IN OPERATION (1917)

SOURCE-B-FIELD-HOSPITAL-XRAY-1917



An RAMC radiographer and medical orderly using a mobile X-ray apparatus powered by a motor vehicle generator to locate metal shrapnel lodged in the skull and chest of a wounded soldier at a Casualty Clearing Station.

Provenance: Official British Army photograph taken in northern France, 1917. Imperial War Museum Collection (Shelfmark: Q 7824).

Context: Official photograph taken in 1917 showing a mobile X-ray unit in operation at an advanced medical station on the Western Front. Radiographers used mobile generators mounted on motor vans to power fluoroscopic screens and photographic plates, allowing surgeons to locate radiopaque shrapnel balls and fractured bone fragments accurately before surgery.

Hinge Question: How does the mobile X-ray setup shown in Source B illustrate how technological advancements directly assisted specialized surgeons like Harvey Cushing and Harold Gillies?

QUESTION 2(A): SINGLE-SOURCE UTILITY ASSESSMENT [4 MARKS • 5 MINS]

EDEXCEL PAPER 1 TARIFF

Study Source A on page 14. How useful is Source A for an enquiry into advances in blood transfusion on the Western Front? [4 marks]

Provenance Clue: Consider the nature of an authentic surviving medical apparatus preserved in the RAMC Museum collection. What physical technology of blood storage in 1917 does it prove, and what clinical administration details does an artifact omit?

Sentence Stems: Source A is useful for investigating blood transfusion advances because it proves... • Robertson's use of sodium citrate and ice allowed blood to be... • However, as a museum artifact, it does not provide clinical data on...

QUESTION 2(B): EDEXCEL FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]

TARGET: SOURCE B

How could you follow up Source B to find out more about treatments or conditions on the Western Front?

Detail in Source B that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

QUESTION 1: FEATURE QUESTIONS [4 MARKS • 5 MINS]
MARK SCHEME: 1 MARK FEATURE + 1 MARK DETAIL
1(a) Describe one feature of the system of communication trenches on the Western Front. [2 marks]

Feature: _____

Detail: _____

1(b) Describe one feature of the treatment for gas gangrene used at Casualty Clearing Stations. [2 marks]

Feature: _____

Detail: _____

QUESTION 2(A): SOURCE UTILITY [8 MARKS • 10 MINS]
SOURCES A AND B ON PAGES 14–15
Study Sources A and B on pages 14–15. How useful are Sources A and B for an enquiry into advances in blood transfusion and surgery on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. [8 marks]

QUESTION 2(B): FOLLOW-UP ENQUIRY TABLE [4 MARKS • 5 MINS]
TARGET: SOURCE B ON PAGE 15
Study Source B on page 15. How could you follow up Source B to find out more about the use of mobile X-ray units to treat wounded soldiers on the Western Front?

Detail in Source B that I would follow up:	_____
Question I would ask:	_____
Type of source I would look for:	_____
How this might help answer my question:	_____

TEACHER ASSESSMENT & PROGRESS TRACKER
Score: ____ / 16 • Grade: ____
What Went Well (WWW):
Even Better If (EBI):

Grade Boundaries: Grade 9: 15–16 | Grade 7: 12–14 | Grade 5: 9–11 | Grade 4: 7–8

Target achieved: [] Yes [] Working Towards